

GARCH Model on Bitcoin Prices

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1 Introduction

Bitcoin and cryptocurrency have always been a blackbox to most of the people. We had always heard about these financial analysts, market researchers and good traders, people want to invest in it for the sake of not missing the hype, but it's definitely one of the hardest markets to invest in for many reasons. Although this field is one of the most studied fields nowadays, it's very hard to predict its prices. A key challenge is the fluctuation of the market and external factors affecting the prices. Sometimes the market is stable and quiet sometimes it's crazy and people start selling their stocks in a state called *Panic Selling* and this features is known as *volatility clustering*.

To deal with this kind of volatility, a model called ARCH was introduced and later extended into a more flexible and general version called GARCH.. These models are now one of the most common and efficient ways to study how the variance of a financial series changes over time.

In this project we study the daily price of Bitcoin (BTC) in US dollars, from April 2013 to May 2026, covering more than thirteen years and 4,763 daily observations. Bitcoin is an interesting asset for this kind of analysis, it's a good example to study because it is well known for being very volatile and for going through several big rises and crashes.

The aim of the project is to investigate whether a GARCH(1,1) can fit and capture the volatility of Bitcoin.

2 Analysis of data

The data used in this project are Bitcoin prices in US dollars at the end of the trading day, downloaded from CoinGecko. The data goes from 28 April 2013 to 14 May 2026 and contains 4,763 daily observations. Figure 1 shows the closing price over the full period.

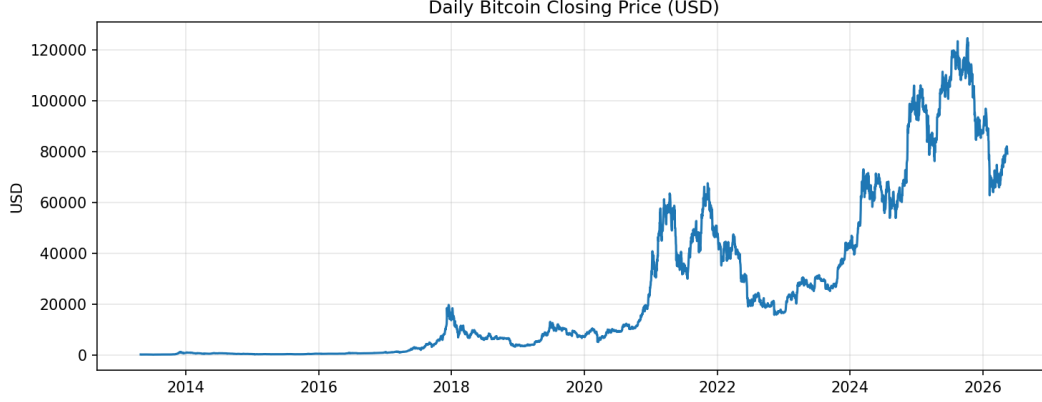


Figure 1: Daily Bitcoin closing price in USD, April 2013 – May 2026.

The price clearly is volatile and does not stay around a constant level. It increases over the years with a lot of big rises and crashes along the way, like the drop in 2018 after its first spike in 2017 and the drop in 2022. In time series terms, this means the series is not stationary, and it cannot be modelled directly, so instead of modeling the prices itself we're gonna model the difference between the prices daily, which is the change percentage per day. We can define it by the coming formula.

$$r_t = \log(P_t) - \log(P_{t-1}), \quad (1)$$

where P_t is the closing price on day t , and P_{t-1} is the closing price on previous day of t . Although Log-returns are almost equal to the daily percentage change, log-returns are preferred in financial time series analysis. Figure 2 shows the log-returns of Bitcoin over the same period.

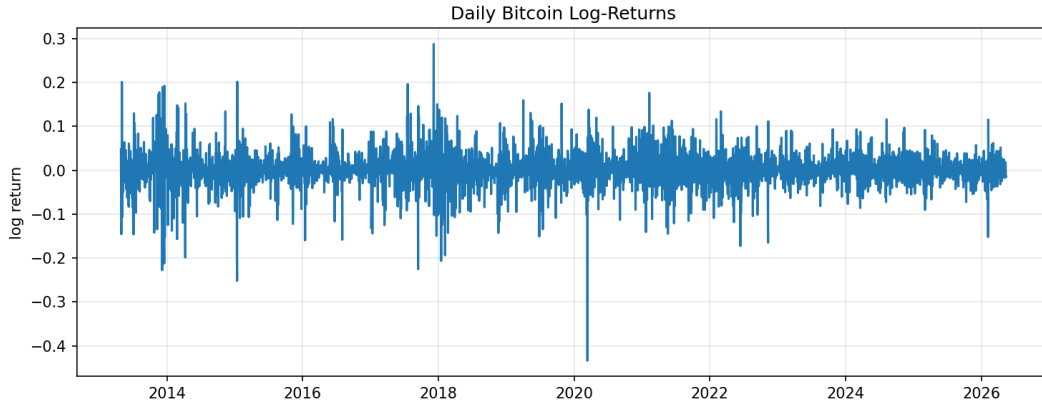


Figure 2: Daily Bitcoin log-returns, April 2013 – May 2026.

The log-returns move around zero with no clear trend, which is a good sign. But the fluctuation of the moves is not constant: some years are quiet and calm such as 2019 or late 2023, while others are very volatile. The biggest drop in the series is around -46% in March 2020, caused by the COVID-19. This behaviour of calmness and turbulence following each other is called volatility clustering pattern mentioned in the introduction, and it is the main reason for using a GARCH model.

To take a closer look at the structure of the log-returns, we plot their ACF and PACF, and also the ACF and PACF of the squared log-returns. The ACF (autocorrelation function) at lag k measures the correlation between today's value and the value of k days ago. The PACF (partial autocorrelation function) measures the same thing, but after removing the effect of the intermediate days, or in other words, finds the direct connection or correlation between today and the value from k ago. Figure 3 shows these four plots together.

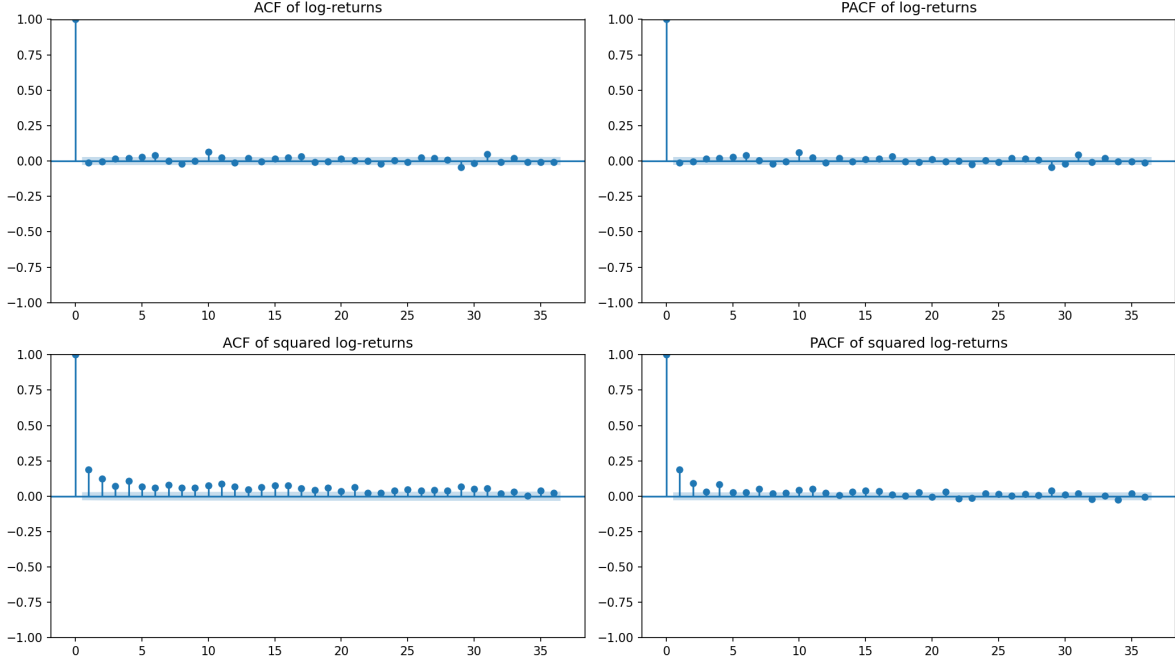


Figure 3: ACF and PACF of log-returns (top row) and squared log-returns (bottom row).

The ACF and PACF of the log-returns stay entirely inside the blue confidence band except for a bar or two who are slightly above the band in random lags. This tells us that today's return is not related to past or lag returns in a useful way. Which means we can't predict if bitcoin's prices will go up or down based on what happened yesterday or in the past few days.

The ACF and PACF of the squared log-returns tell a different story. The ACF shows positive bars which are slowly decaying across the lags, also the PACF has clear positive bars at the beginning. This means that although the returns are not predictable but the size has a memory. We can see that if a day has a big move most probably the next day will have a big move too and if the day was calm it's high chance that the next day will be also calm.

To confirm this, we run the Ljung-Box test at lags 5, 10 and 20 to see if there is enough evidence to say the series is correlated or not. A small p-value means lower chances to find correlation. The results for both the log-returns and the squared log-returns are shown in Table 1.

Lag	p-value (returns)	p-value (squared returns)
5	0.178	$< 10^{-71}$
10	4.1×10^{-5}	$< 10^{-89}$
20	6.4×10^{-5}	$< 10^{-122}$

Table 1: Ljung-Box test p-values on log-returns and squared log-returns.

For the squared log-returns the p-values are extremely small at every lag, which means that the squared returns are autocorrelated. This is a proof that the variance has memory, and that's the reason why we are using GARCH.

For the log-returns the p-value at lag 5 is 0.178, which is large enough to say the returns are uncorrelated. At lags 10 and 20 the p-values is getting smaller, but if we look back at the ACF plot, all the bars there are tiny ($\ll 0.05$). Because our sample has more than 4,700 observations it detects a correlations statistically significant.

In short, the analysis shows that we should model the log-returns with a constant mean and a variance that changes over time, which is exactly what a GARCH model does.

For the squared log-returns the p-values are extremely small at every lag, which is strong evidence that the squared returns are autocorrelated. This confirms that there are real ARCH effects in the data, just as the ACF plot suggested.

The conclusion is we should model the log-returns with a constant mean but for the variance we are using GARCH model.

the conclusion: The mean of returns doesn't need a fancy model (constant is fine). The variance does need a fancy model (GARCH).

3 GARCH model

Based on the analysis in the previous section, we fit a GARCH(1,1) model with a constant mean and a Student-t distribution to the log-returns. The model is written as:

$$r_t = \mu + a_t, \tag{2}$$

$$a_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \sim t_\nu, \tag{3}$$

$$\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2. \tag{4}$$

For the first equation means today's return equals to constant average plus error. Here r_t is the log-return on day t , μ is the mean return, a_t is the error on day t . The second equation means the random error of consists of two parts first is today's volatility multiplied by random number. σ_t is today's standard deviation(it says how big the move on day t is), and ε_t is a random number comes from a Student-t distribution. Equation three means today's variance depends on yesterday's squared error and yesterday's variance.

The first line says today's return is the average plus a random error. The second line says the size of the error depends on today's volatility σ_t times a random number. The third line is the main part of the model: today's variance depends on yesterday's squared error and yesterday's variance.

Since bitcoin has extreme volatile and change in prices so the normal distibtion wouldn't capture that so we don't use normal distribution. Instead of that we're using Student-t distribution. The Student-t distribution has an extra parameter ν that controls how heavy the tails are. And since bitcoin has these extreme values so we're expecting a smaller values of ν .

The five parameters μ , ω , α , β , and ν are estimated by maximum likelihood. We split the data into two parts: a training set (4,642 daily observations), used to fit the model, and a test set of 120 observations to evaluate the performance of the model.

The parameters are reported in Table 2.

Parameter	Estimate	Std. Error	<i>t</i> -statistic	<i>p</i> -value
μ	0.1228	0.0297	4.14	3.6×10^{-5}
ω	0.2999	0.1010	2.97	3.0×10^{-3}
α	0.1300	0.0150	8.65	5.1×10^{-18}
β	0.8700	0.0191	45.61	≈ 0
ν	3.1200	0.1120	27.88	$< 10^{-170}$

Table 2: Estimated parameters of the GARCH(1,1) model with Student-t distribution.

All five parameters are significant. The fitted model is therefore:

$$r_t = 0.1228 + a_t, \quad (5)$$

$$a_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \sim t_{3.12}, \quad (6)$$

$$\sigma_t^2 = 0.300 + 0.130 a_{t-1}^2 + 0.870 \sigma_{t-1}^2. \quad (7)$$

The log-likelihood is $-11,633.3$.

Since the $\nu = 3.12$ (very low) it means that using the t-distribution was better as bitcoin has extreme moves often more than the normal distribution would allow.

Evaluation

To check whether the model actually fits the data well, we look at the standardized residuals. If the model is correct then they should look like random noise with no pattern.

Figure 4 shows the ACFs of the standardised residuals and the squared standardised residuals.

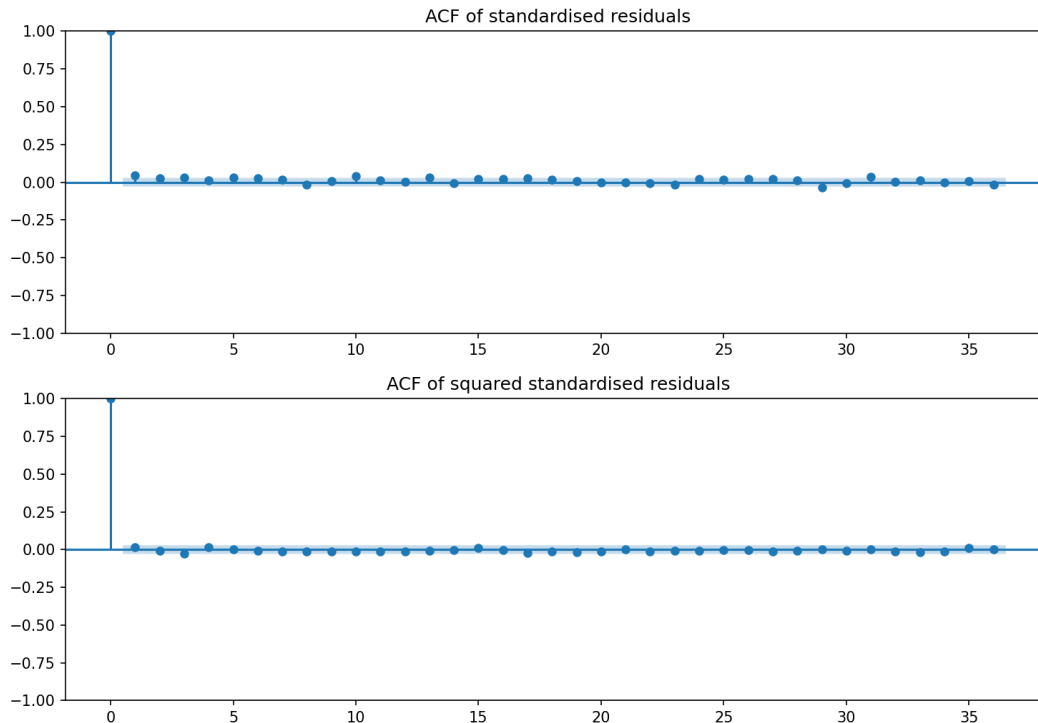


Figure 4: ACF of standardised residuals (top) and squared standardised residuals (bottom).

Both ACF of standardized residuals and the ACF of squared standardized residuals are now mostly inside the confidence band.

To confirm this formally, we run the Ljung-Box test on the standardised residuals and their squares.

Lag	p-value (residuals)	p-value (squared residuals)
5	0.0004	0.294
10	4.7×10^{-5}	0.514
20	0.0001	0.704

Table 3: Ljung-Box test on standardised residuals and squared standardised residuals.

The p-values for the squared residuals are all large (> 0.05) unlike the values before fitting which were very small, which then shows a great evidence that there is no autocorrelation anymore. The p-values for the standardised residuals themselves are small at all three lags, but as discussed in Section 2, this is caused by the very large sample size making the test detect even very small correlations. The actual values are below 0.05. The model works, the standardized residuals behave well, and the squared standardized residuals have lost the autocorrelation pattern they had before fitting. The model is then a good fit.

4 Forecast

We now use the fitted model to forecast volatility for test period, from 15 January 2026 to 14 May 2026 (120 daily observations). The forecasting method is a one-step-ahead rolling forecast because we move forward day by day and we predict only the next day not far in the future. We don't refit the model every day we refit every 20 days as fitting the model takes time and the parameters don't change that much from day to another so it's a waste of power if we refit everyday

Figure 5 compares the forecasted volatility σ_t with the absolute log-returns over the test period.

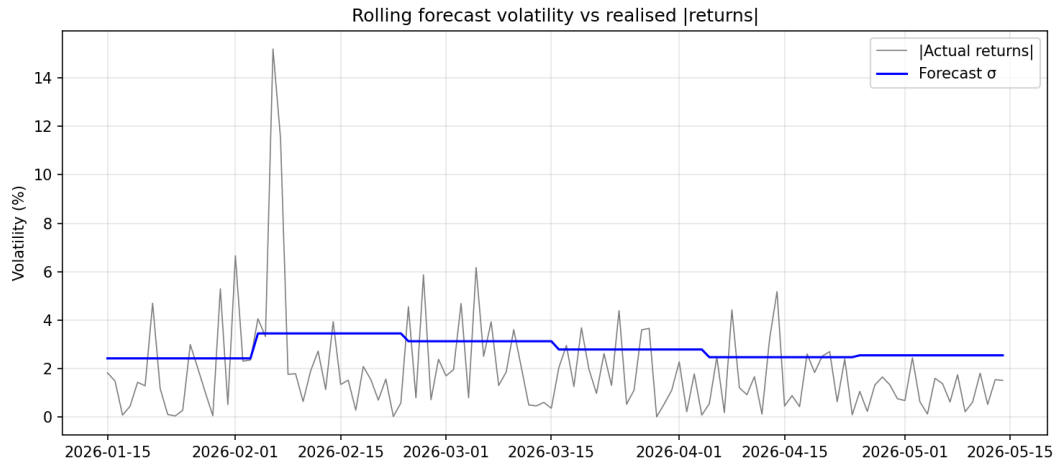


Figure 5: Rolling one-step-ahead forecast volatility (blue) and absolute log-returns (grey) over the test period.

The forecast volatility follows a general pattern. The blue line stays around 2.5% most of the time. In early February 2026, there's a big move in the gray line, a day with about a +15% return. Although the model didn't predict this but after that the blue line goes from 2.5% to about 3.5%, which means the model responds to the spikes. Then over the next period as the market calms down so the blue line slowly kept its value around 2.5%. This is the expected behaviour of GARCH models as they cannot predict a severe spike or an intense drop, but once this occurs they respond.

5 Conclusions

In this project we fitted a GARCH(1,1) model with a t-distribution to 13 years of daily Bitcoin log-returns. The analysis of the data showed that the returns themselves are uncorrelated, but their squares are autocorrelated which is the typical signature of changing variance and the main reason for using a GARCH model. Residual diagnostics confirmed that the model captured the variance structure well and the squared standardized residuals didn't show any signs of autocorrelation anymore. Finally, the rolling forecast on the testing period tracked the volatility, responding to spikes and drops during the whole period. While GARCH models cannot predict spikes, they remain a useful tool for describing and forecasting the volatility of Bitcoin returns.